3种方法

1. use circle

$$A=\int \pi r^2\,dx \quad \text{or} \quad A=\int \pi r^2\,dy$$

⇓ 中间 hollow

$$A=\int \pi r_{out}^2-\pi r_{in}^2\,dx \quad \text{or} \quad \int \cdots\,dy$$

2. use cross section

$$A=\int A(x)\,dx$$

dx → 截面 → square: $A(x)=x^2$, triangle.

3. use cylinder

$A = \int 2\pi r h \, dx$ or $A = \int 2\pi r h \, dy$

CHAPTER 6 APPLICATIONS OF INTEGRATION

6.1 Areas between Curves

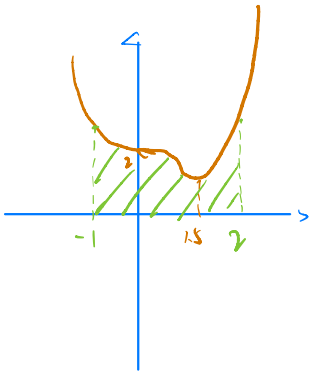
- **Region Above the x-Axis**

Let $y = f(x)$ be continuous and nonnegative on $a \le x \le b$, and let R be the region bounded by the curve $y = f(x)$, the lines $x = a$, $x = b$ and the x axis. The area of the region R is

$$A = \int_a^b f(x)dx$$

Example 6.1. Find the area of the region R under $y = x^4 - 2x^3 + 2$ between $x = -1$ and $x = 2$.

graph will be given



$$A = \int_{-1}^{2} x^4 - 2x^3 + 2\,dx = \left[\frac{x^5}{5} - 2x^4 + 2x\right]_{-1}^{2}$$

$$= \frac{51}{10}$$

- **Region Below the x-Axis**

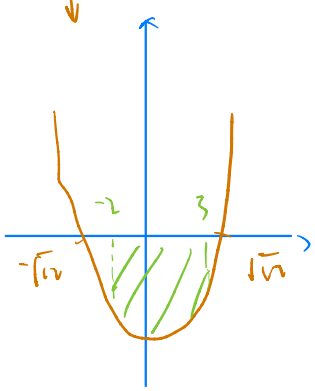
Let $y = f(x)$ be continuous and non-positive on $a \leq x \leq b$, and let R be the region bounded by the curve $y = f(x)$, the lines $x = a$, $x = b$ and the x axis. The area of the region R is

$$A = -\int_a^b f(x)dx$$

Example 6.2. Find the area of the region R bounded by $y = \frac{x^2}{3} - 4$, the x axis, $x = -2$ and $x = 3$.

area below the curve

will be given



$$A = -\int_{-2}^{3} \frac{x^2}{3} - 4 \, dx$$

$$= -\left[\frac{x^3}{9} - 4x\right]_{-2}^{3}$$

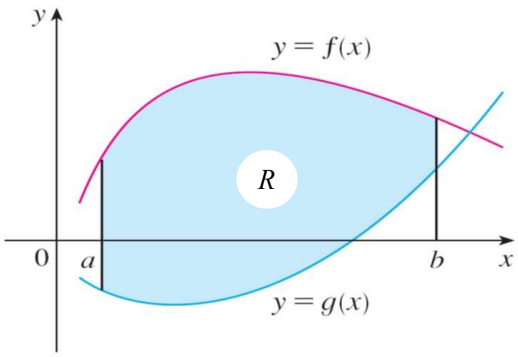
$$= \frac{125}{9}$$

- **A Region Between Two Curves**

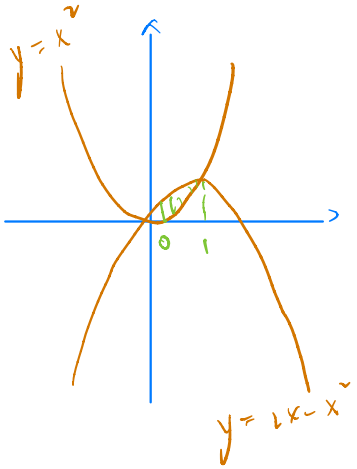
△

Consider the curves $y = f(x)$ and $y = g(x)$ with $g(x) \le f(x)$ on $a \le x \le b$. Let R be the region bounded between the curves $y = f(x)$, $y = g(x)$ and the lines $x = a$ and $x = b$. The area of R is

$$A = \int_a^b [f(x) - g(x)]\, dx$$



Example 6.3. Find the area of the region between the curves $y = x^2$ and $y = 2x - x^2$.



Intersection point?

$x^2 = 2x - x^2$

$2x^2 - 2x = 0$

$2x(x-1) = 0$

$x = 0$ or $x = 1$

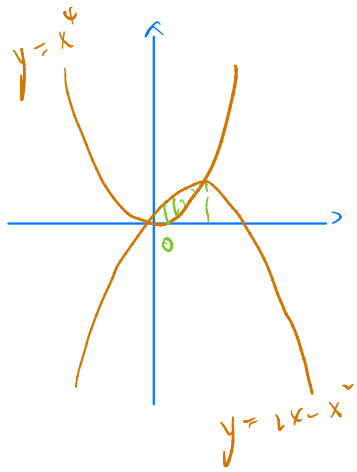
$$A = \int_0^1 [(2x - x^2) - x^2]\, dx$$

$$= \int_0^1 2x - 2x^2\, dx$$

$$= 2\int_0^1 x - x^2\, dx$$

$$= 2\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = 2\left[\frac{1}{2} - \frac{1}{3}\right] = \frac{1}{3}$$

Example 6.4. Find the area of the region between the curves $y = x^4$ and $y = 2x - x^2$.



intersection:

$x^4 = 2x - x^2$

$x^4 + x^2 - 2x = 0$

$x(x^3 + x - 2) = 0$

$x = 0$ 试根: when $x = 1$, $x^3 + x - 2 = 0$ → $(x-1)(\quad)$

$x(x-1)(x^2+x+2) = 0$ 然后用 Long Division 找另一个根.

$x = 0$ $x = 1$ 无 (no real roots)

Long Division

$$\begin{array}{r} x^2 + x + 2 \\ x-1 \overline{)\, x^3 + 0x^2 + x - 2} \\ \underline{x^3 - x^2} \\ +x^2 + x \\ \underline{x^2 - x} \\ 2x - 2 \\ \underline{2x - 2} \\ 0 \end{array}$$

do subtraction here: $x^3 - x^3 = 0$, $0x^2 - (-x^2) = 2x^2$

$$A = \int_0^1 [(2x - x^2) - x^4]\,dx$$

$$= \int_0^1 2x - x^2 - x^4\,dx$$

$$= \left[x^2 - \frac{x^3}{3} - \frac{x^5}{5}\right]_0^1$$

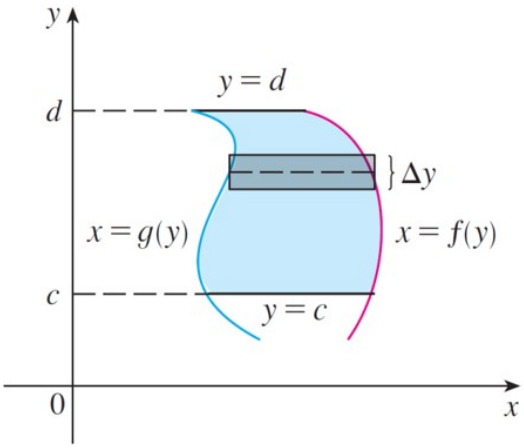
$$= \left(1 - \frac{1}{3} - \frac{1}{5}\right) =$$

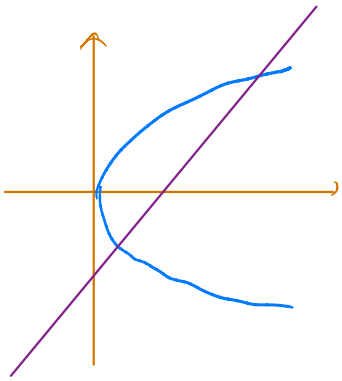
- **Horizontal Slicing**

Consider the curves $x = f(y)$ and $x = g(y)$ with $g(y) \le f(y)$ on $c \le y \le d$. Let R be the region bounded between the curves $x = f(y)$, $x = g(y)$ and the lines $y = c$ and $y = d$. The area of R is

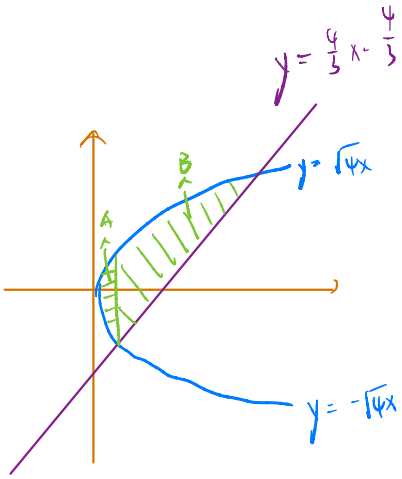
$$A = \int_c^d [f(y) - g(y)]\, dy$$

right left





Example 6.5. Find the area of the region bounded between the parabola $y^2 = 4x$ and the line $4x - 3y = 4$.



use vertical integration

Area A

Area B

∵ 有兩個 Cover function

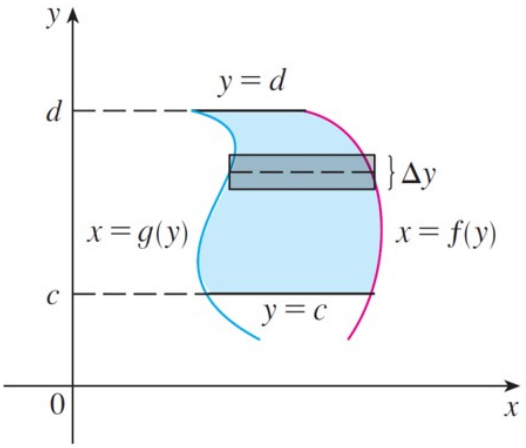
$y = -\sqrt{4x}$ Region A

$y = \frac{4}{3}x - \frac{4}{3}$ Region B

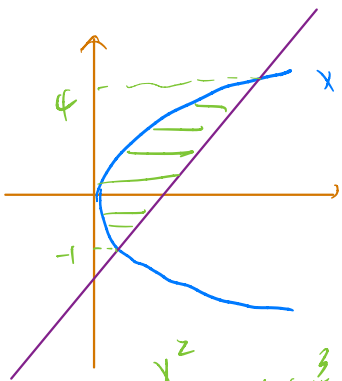
- **Horizontal Slicing**

Consider the curves $x = f(y)$ and $x = g(y)$ with $g(y) \le f(y)$ on $c \le y \le d$. Let R be the region bounded between the curves $x = f(y)$, $x = g(y)$ and the lines $y = c$ and $y = d$. The area of R is

$$A = \int_c^d [f(y) - g(y)]\, dy$$



Example 6.5. Find the area of the region bounded between the parabola $y^2 = 4x$ and the line $4x - 3y = 4$.



$x = 1 + \frac{3}{4}y$

$f_R - f_L$.

$$\int_{-1}^{4} \left[\frac{y^2}{4} - \left(1 + \frac{3}{4}y\right)\right] dx$$

$$= \left[\frac{y^3}{12} - y + \frac{3}{8}y^2\right]_{-1}^{4}$$

$$= \int_{-1}^{4} 1 + \frac{3}{4}y - \frac{y^2}{4}\, dx$$

$$= \left[y + \frac{3}{8}y^2 - \frac{y^3}{12}\right]_{-1}^{4}$$

$$= \frac{125}{24}$$

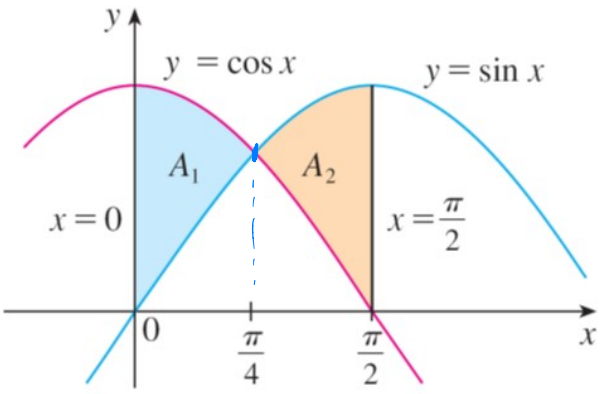
$\frac{y^2}{4} = 1 + \frac{3}{4}y$

$y^2 - 3y - 4 = 0$

$(y-4)(y+1) = 0$

$y = 4$ or $y = -1$

Example 6.6. Find the area of the region bounded by the curves $y = \sin x$, $y = \cos x$, $x = 0$ and $x = \dfrac{\pi}{2}$.



Use vertical integration - x region

(If horizontal integration - y region)

① find intersection point

$\sin x = \cos x$

$\tan x = 1$

$x = \frac{\pi}{4}$

② vertical integration

$$\text{Area} = \int_0^{\frac{\pi}{4}} \cos x - \sin x \, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin x - \cos x \, dx$$

$$= \left[\sin x + \cos x \right]_0^{\frac{\pi}{4}} + \left[-\cos x - \sin x \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \sqrt{2} - 1$$

Exercises 6.1 1, 3, 7, 13, 15, 17, 21, 25, 27, 57, 61.

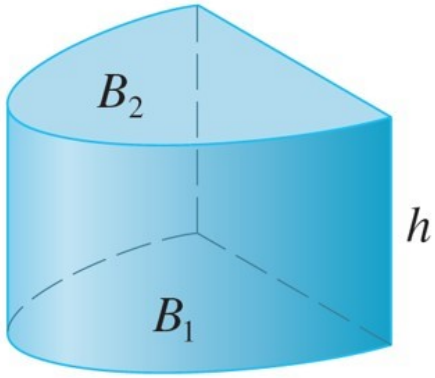
6.2 Volumes

cross - sectional Area

- The volume of a prism is

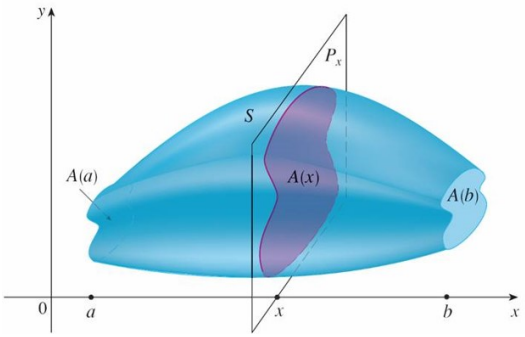
$$V = A \cdot h$$

where A is the area of the base and h is the height.



- For a general solid,

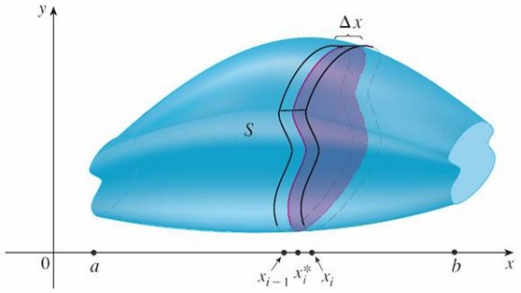
$$V = \int_a^b A(x)dx$$

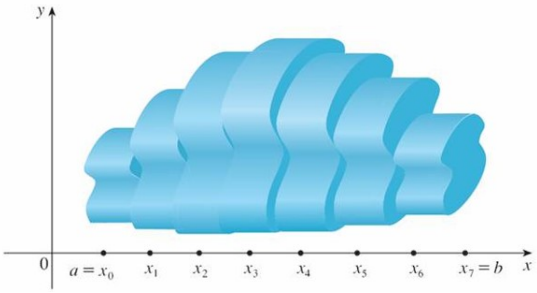


$V = \sum_{i=1}^{\infty} A(x_i)\,\Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i)\Delta x = \int_a^b A(x)\,dx$

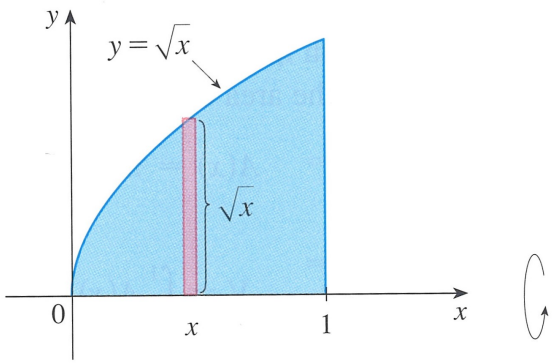
Riemen Sum



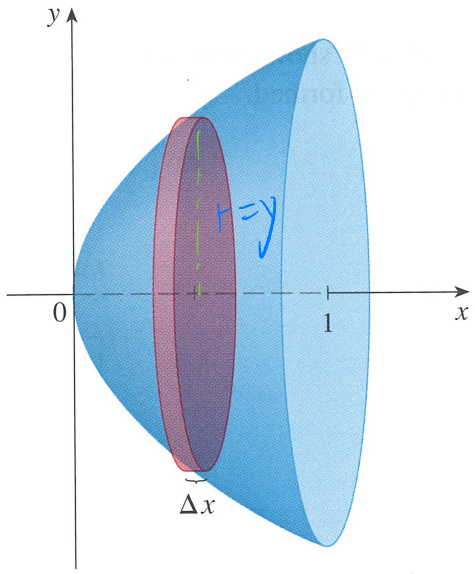


- When a plane region, lying entirely on one side of a fixed line in its plane, is revolved about that line, it generates a solid of revolution.

Revolve about x-axis



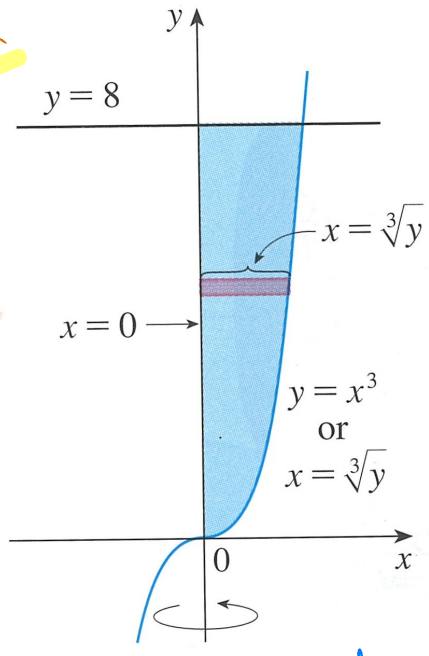
Disk

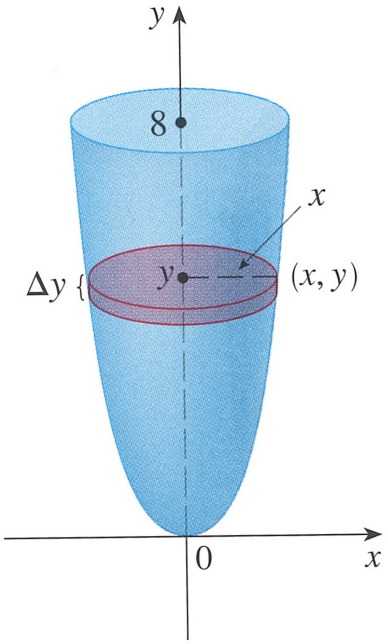


Area of base of disk = πr^2

Volume = $\int_a^b \pi r^2 \cdot dx = \int_a^b \pi y^2 dx$

Revolve about y-axis





Volume = $\int_c^d \pi x^2 dy$

Example 6.7. Find the volume of the solid of revolution obtained by revolving the plane region R bounded by $y = \sqrt{x}$, the x-axis, and the line $x = 1$ about the x-axis.

$$\text{Volume} = \int_0^1 \pi y^2 \, dx = \int_0^1 \pi (\sqrt{x})^2 \, dx$$

$$= \int_0^1 \pi x \, dx$$

$$= \left[\frac{\pi x^2}{2} \right]_0^1 =$$

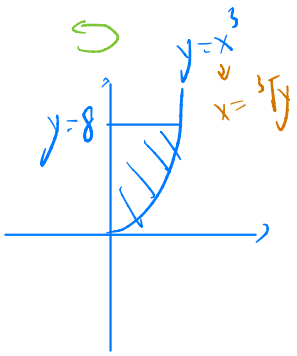
Example 6.8. Find the volume of the solid generated by revolving the region bounded by the curve $y = x^3$, the y-axis and the line $y = 8$ about the y-axis.

$x = \sqrt[3]{y}$

$$\text{Volume} = \int_0^8 \pi x^2 \, dy$$

$$= \int_0^8 \pi (\sqrt[3]{y})^2 \, dy$$

$$= \frac{3}{5} \pi \left[y^{\frac{5}{3}} \right]_0^8 = \frac{3}{5} \pi \cdot \frac{96}{5} \pi$$



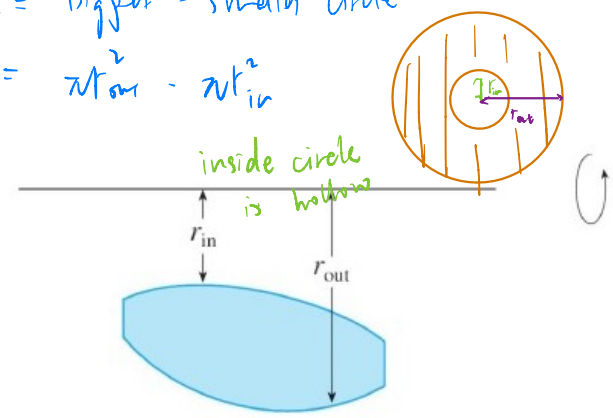
Washer method

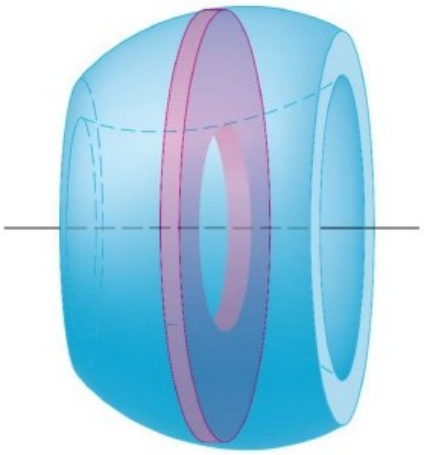
- Sometimes, slicing a solid of revolution results in disks with holes in the middle. We call them washers.

Single layer

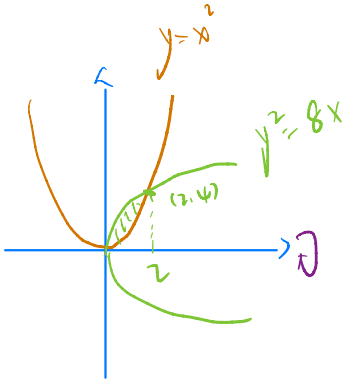
Area = bigger - smaller circle

$= \pi r_{out}^2 - \pi r_{in}^2$





Example 6.9. Find the volume of the solid generated by revolving the region bounded by the parabola $y = x^2$ and $y^2 = 8x$ about the x-axis.



$$\text{Area} = \int_a^b \pi r_{out}^2 - \pi r_{in}^2 \, dx$$

$$= \int_0^2 \pi \cdot (\sqrt{8x})^2 - \pi (x^2)^2 \, dx$$

$$= \pi \int_0^2 8x - x^4 \, dx$$

intersect point

$x^2 = \sqrt{8x}$

$x^4 = 8x$

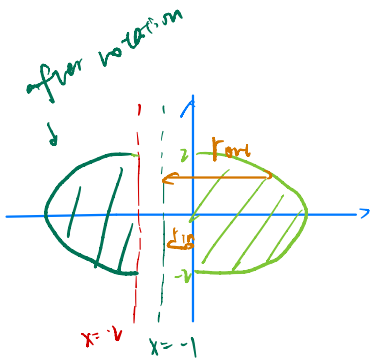
$x(x^3 - 8) = 0$

$x = 0$ or 2

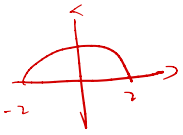
$=$

$= \frac{48}{5}\pi$

Example 6.10. The semicircular region bounded by the curve $x = \sqrt{4 - y^2}$ and the y-axis is revolved about the line $x = -1$. Set up the integral that represents its volume. Then compute this volume.



$r_{out} = \sqrt{4-y^2} + 1$; $r_{in} = 1$



Semicircle with $r = 2$

$$\text{Area} = \int_c^d \pi r_{out}^2 - \pi r_{in}^2 \, dy$$

$$= \pi \int_{-2}^{2} (\sqrt{4-y^2}+1)^2 - 1^2 \, dy$$

$$= \pi \int_{-2}^{2} 4 - y^2 + 2\sqrt{4-y^2} \, dy$$

$$= \pi \left[4y - \frac{y^3}{3} + \frac{4}{3}(4-y^2)^{\frac{3}{2}} \right]_{-2}^{2}$$

$$= \pi \left[\left(4\times 2 - \frac{8}{3} + \frac{4}{3}(4-4)^{\frac{3}{2}}\right) - \left(4\times(-2) + \frac{8}{3} + \frac{4}{3}(4-4)^{\frac{3}{2}}\right) \right]$$

$$= \pi \times 10.667$$

$$= 33.51 \text{ u}^3$$

$$2\int_{-2}^{2} \sqrt{4-y^2} \, dy = 2 \times \frac{1}{2}\pi \times 2^2$$

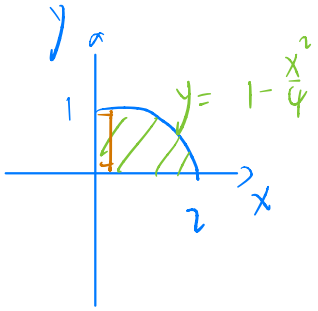
$$= 4\pi$$

$$=$$

Let $u = 4 - y^2$ $\quad \frac{du}{dy} = -2y$

$$\int_{-2}^{2} \sqrt{4-y^2} \, dy$$

Example 6.11. Let the base of a solid be the first quadrant plane region bounded by $y = 1 - \frac{x^2}{4}$, the x-axis, and the y-axis. Suppose that the cross sections perpendicular to the x-axis are squares. Find the volume of the solid.



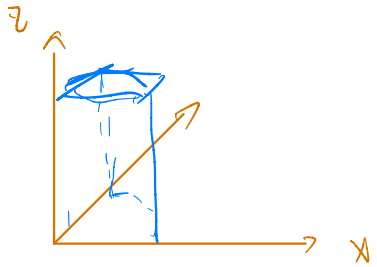
$$V = \int_a^b A(x)\,dx$$

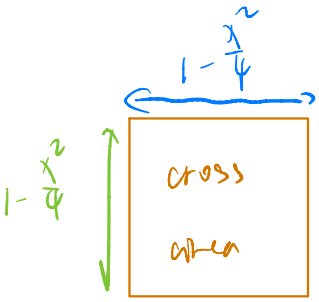
$$= \int_0^2 \left(1 - \frac{x^2}{4}\right)^2 dx$$

$$= \int_0^2 1 - \frac{x^2}{2} + \frac{x^4}{16}\,dx$$

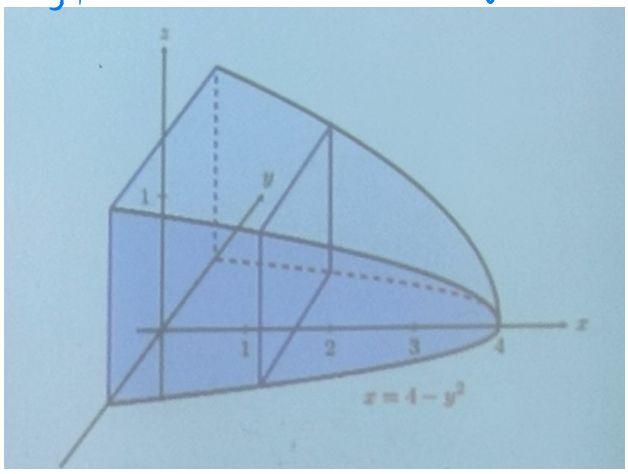
$$= \left[x - \frac{x^3}{6} + \frac{x^5}{80}\right]_0^2$$

$$= \frac{16}{15}$$

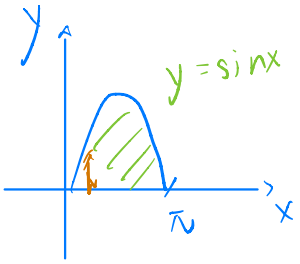




similar to ↓



Example 6.12. The base of a solid is the region between one arch of $y = \sin x$ and the x-axis. Each cross section perpendicular to the x-axis is an equilateral triangle sitting on its base. Find the volume of the solid.



$$V = \int_a^b A(x)\,dx$$

$$= \int_0^{\pi} \frac{\sqrt{3}\sin^2 x}{4}\,dx$$

$$= \frac{\sqrt{3}}{4} \cdot \int_0^{\pi} \sin^2 x\,dx$$

reduce to power 1

$\cos 2A = \cos^2 A - \sin^2 A$

$= 2\cos^2 A - 1$

$= 1 - 2\sin^2 A$

$\sin^2 A = \frac{1 - \cos 2A}{2}$

$$= \frac{\sqrt{3}}{4} \cdot \left[\frac{\sin^3 x}{3\cos x}\right]_0^{\pi}$$

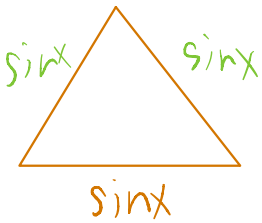
$$= \frac{\sqrt{3}}{12}\left(\frac{\sin^3 \pi}{\cos \pi} - \frac{\sin^3 0}{\cos 0}\right)$$

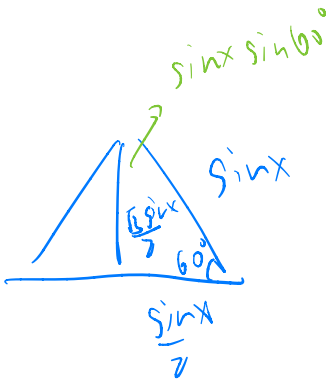
$$= \frac{\sqrt{3}}{12}(0 - 0) = 0$$

$$= \frac{\sqrt{3}}{4}\int_0^{\pi} \frac{1}{2} - \frac{\cos 2A}{2}\,dx$$

$$= \frac{\sqrt{3}}{4}\left[\frac{x}{2}\right]_0^{\pi} - \frac{3}{16}\left[\sin 2A\right]_0^{\pi}$$

$$= \frac{\sqrt{3}}{4} \times \frac{\pi}{2} = \frac{\sqrt{3}}{8}\pi$$





$\cos^2 2A = \cos^2 A - \sin^2 A$

$= 2\cos^2 A - 1$

$= 1 - 2\sin^2 A$

Exercises 6.2 5, 7, 9, 11, 13, 15, 17, 47, 49, 53, 55, 63.

6.3 Volumes by Cylindrical Shells

- A cylindrical shell is a solid bounded by two concentric right circular cylinders. If the inner radius is r_1 and the outer radius is r_2, and the height is h, then the volume of the cylindrical shell is

larger cylinder — Volume of cylinder : $V_c = \pi r^2 h$

$$V = \pi \left(r_2^2 - r_1^2\right) h$$

smaller cylinder

$= \pi (r_2 + r_1)(r_2 - r_1) h$

let $r = \frac{r_1 + r_2}{2}$; $\Delta r = r_2 - r_1$

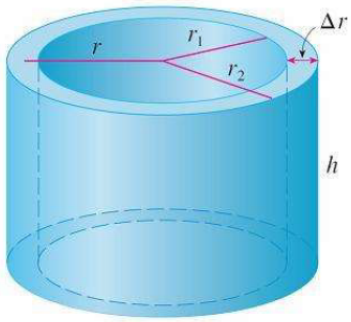
$= \pi \cdot 2 \left(\frac{r_1 + r_2}{2}\right) (r_2 - r_1) h$

$= \pi \cdot 2 r \cdot \Delta r \cdot h$

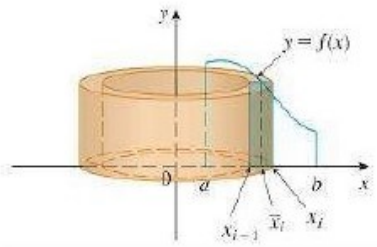
$= 2 \pi r h \Delta r$ — consider $r_1 \simeq r_2$, $\Delta r \to 0$

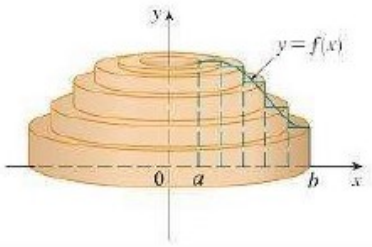
$= 2\pi r h$

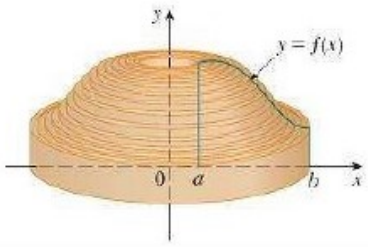
$r = \frac{r_1 + r_2}{2}$



$\Delta r = r_2 - r_1$



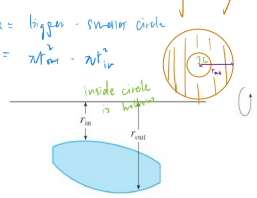


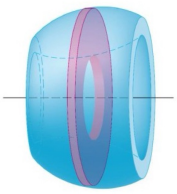


- Sometimes, slicing a solid of revolution results in disks with holes in the middle. We call them washers.

Single layer

Area = bigger circle − smaller circle $= \pi r_{out}^2 - \pi r_{in}^2$

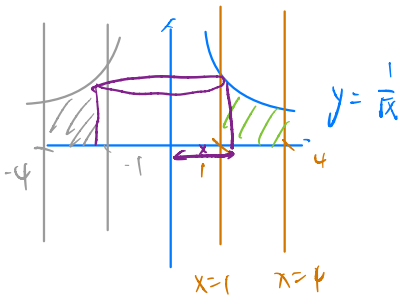




⇒ 如左行不同於 integration 的方式不好

⇒ shallow cylinder

Example 6.13. The region bounded by $y = \dfrac{1}{\sqrt{x}}$, the x-axis, $x = 1$ and $x = 4$ is revolved about the y-axis. Find the volume of the resulting solid.



$$V = \int_0^1 A(x)\,dx$$

$$= \int_1^4 2\pi r h\,dx$$

$$= \int_1^4 2\pi \cdot x \cdot \frac{1}{\sqrt{x}}\,dx$$

$$= 2\pi \int_1^4 \sqrt{x}\,dx$$

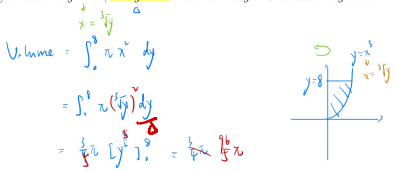
$$= \frac{4}{3}\pi \left[x^{\frac{3}{2}}\right]_1^4$$

$$= \frac{4}{3}\pi (8 - 1)$$

$$= \frac{28}{3}\pi$$

washer method : integration about y

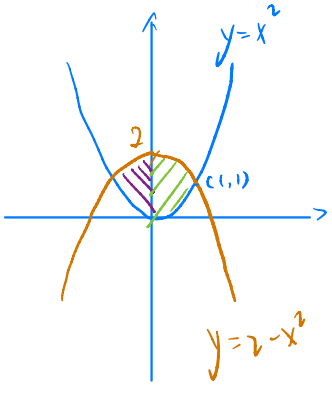
Example 6.8. Find the volume of the solid generated by revolving the region bounded by the curve $y = x^3$, the y-axis and the line $y = 8$ about the y-axis.



Example 6.14. Find the volume of the solid generated by revolving the region in the first quadrant that is above the parabola $y = x^2$ and below the parabola $y = 2 - x^2$ about the y-axis.

x, $(2-x^2)-x^2 \to (2-x^2)-x^2$

$$V = \int_0^1 2\pi r h\,dx$$

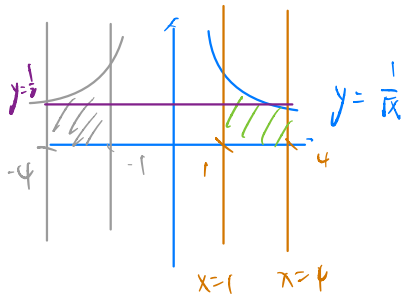


$$= \int_0^1 2\pi x\left[(2-x^2)-x^2\right]dx$$

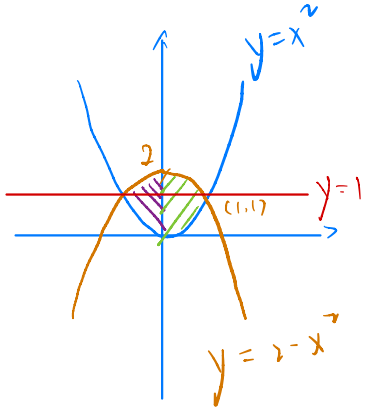
$$= 2\pi \int_0^1 x(2-2x^2)\,dx$$



Example 6.13. The region bounded by $y = \dfrac{1}{\sqrt{x}}$, the x-axis, $x = 1$ and $x = 4$ is revolved about the y-axis. Find the volume of the resulting solid.



Example 6.14. Find the volume of the solid generated by revolving the region in the first quadrant that is above the parabola $y = x^2$ and below the parabola $y = 2 - x^2$ about the y-axis.



try this

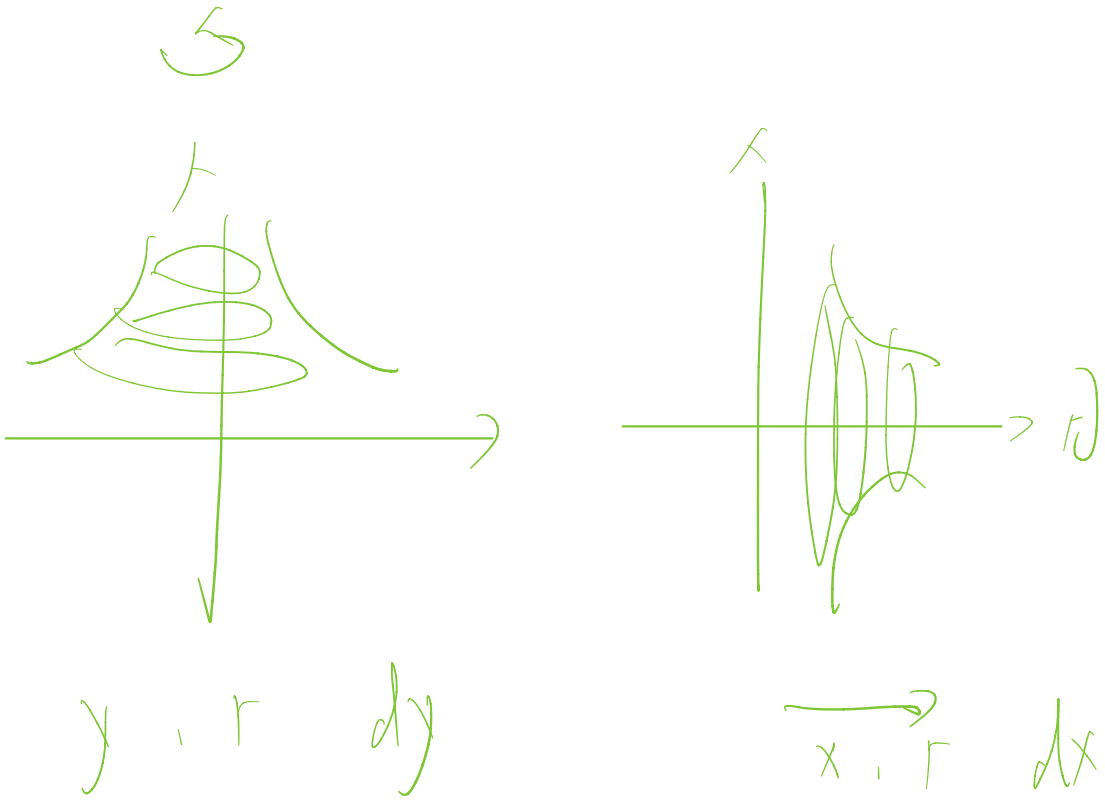
Example 6.15. Set up and evaluate an integral for the volume of the solid that results when the region R bounded by the curve $y = 3 + 2x - x^2$ and the two coordinate axes in the first quadrant is revolved about

(a) the x-axis, Disc

(b) the y-axis, shell

(c) the line $y = -1$, Disc

(d) the line $x = 4$. shell



Exercises 6.3 5, 7, 11, 13, 15, 17, 19, 25(a), 39, 41, 43.

Example 6.15. Set up and evaluate an integral for the volume of the solid that results when the region R bounded by the curve $y = 3 + 2x - x^2$ and the two coordinate axes in the first quadrant is revolved about

(a) the x-axis,

(b) the y-axis,

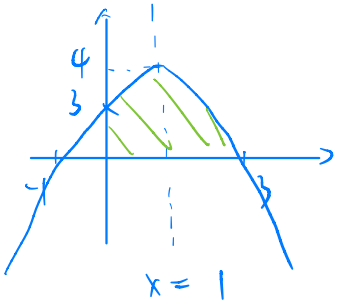
(c) the line $y = -1$,

(d) the line $x = 4$.

$-x^2 + 2x = y - 3$

$-(x-1)^2 = y - 4$

a)



$V = \int_0^3 2\pi r h \, dy$

$= \int_0^3 2\pi \cdot 2y \cdot -$



$r = y$

$h =$



Exercises 6.3 5, 7, 11, 13, 15, 17, 19, 25(a), 39, 41, 43.

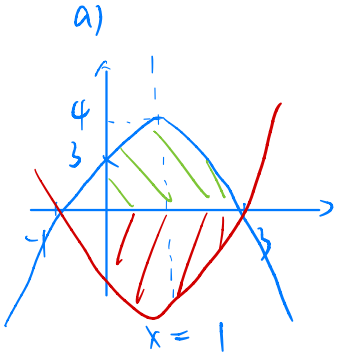
Example 6.15. Set up and evaluate an integral for the volume of the solid that results when the region R bounded by the curve $y = 3 + 2x - x^2$ and the two coordinate axes in the first quadrant is revolved about

(a) the x-axis,

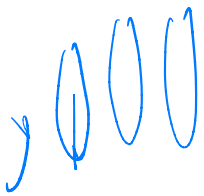
(b) the y-axis,

(c) the line $y = -1$,

(d) the line $x = 4$.



$y = 3 + 2x - x^2$



$(a+b)^2 = a^2 + 2ab + b^2$

$$\int_0^3 \pi y^2\,dx = \int_0^3 \pi (3+2x-x^2)^2\,dx$$

$$= \pi \int_0^3 9 + 4x^2 - x^4\,dx$$

$$= \pi \left[9x + \frac{4}{3}x^3 - \frac{1}{5}x^5\right]_0^3$$

$$= \pi \left(9\times 3 + 4\times 3^2 - \frac{1}{5}\times 3^5\right)$$

$$= \frac{72}{5}\pi$$

$$= \pi \int_0^3 (9 - 2x^2 - 4x^3 + x^4)\,dx$$

$$= \pi \left[9x - \frac{2}{3}x^3 - x^4 + \frac{1}{5}x^5\right]_0^3$$

$$= \pi \left(9\times 3 - \frac{2}{3}\times 3^3 - 3^4 + \frac{1}{5}\times 3^5\right)$$

$6x^2$

6×3^2

$$= \frac{108}{5}\pi$$

$(3+2x-x^2)(3+2x-x^2)$

$$= 9 + 6x - 3x^2 + 6x + 4x^2 - 2x^3 - 3x^2 - 2x^3 + x^4$$

$$= 9 - 2x^2 - 4x^3 + x^4 + 12x$$

Exercises 6.3 5, 7, 11, 13, 15, 17, 19, 25(a), 39, 41, 43.

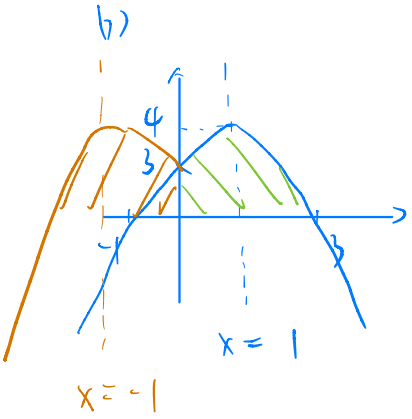
Example 6.15. Set up and evaluate an integral for the volume of the solid that results when the region R bounded by the curve $y = 3 + 2x - x^2$ and the two coordinate axes in the first quadrant is revolved about

(a) the x-axis,

(b) the y-axis, shell

(c) the line $y = -1$,

(d) the line $x = 4$.



$$V = \int_0^3 A(x)\,dx$$

$$= \int_0^3 2\pi \cdot x \cdot h\,dx \qquad x \to x,\quad y \to y = 3+2x-x^2$$

$$= \int_0^3 2\pi \cdot x\,(3+2x-x^2)\,dx$$

$$= 2\pi \int_0^3 3x + 2x^2 - x^3\,dx$$

$$= 2\pi \left[\frac{3}{2}x^2 + \frac{2}{3}x^3 - \frac{1}{4}x^4\right]_0^3$$

$$= 2\pi \times \frac{45}{4}$$

$$= \frac{45}{2}\pi$$

Exercises 6.3 5, 7, 11, 13, 15, 17, 19, 25(a), 39, 41, 43.

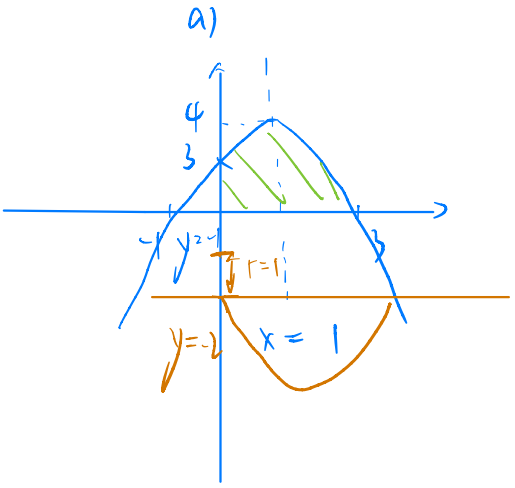
Example 6.15. Set up and evaluate an integral for the volume of the solid that results when the region R bounded by the curve $y = 3 + 2x - x^2$ and the two coordinate axes in the first quadrant is revolved about

(a) the x-axis,

(b) the y-axis,

(c) the line $y = -1$, washer

(d) the line $x = 4$.



$$V = \int_0^4 \pi r^2 - \pi \cdot 1^2 \, dx \qquad (y+1)$$

$$= \int_0^4 \pi \cdot (3 + 2x - x^2) - \pi \, dx$$

$$= \int_0^4 \pi (4 + 2x - x^2)^2 - \pi \, dx$$

Exercises 6.3 5, 7, 11, 13, 15, 17, 19, 25(a), 39, 41, 43.

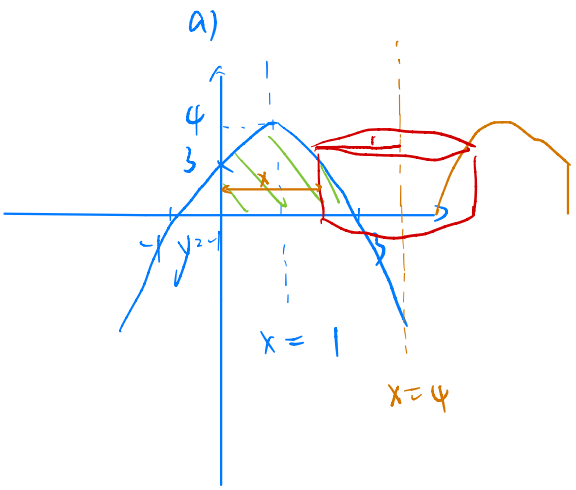
Example 6.15. Set up and evaluate an integral for the volume of the solid that results when the region R bounded by the curve $y = 3 + 2x - x^2$ and the two coordinate axes in the first quadrant is revolved about

(a) the x-axis,

(b) the y-axis,

(c) the line $y = -1$,

(d) the line $x = 4$.



$$A = \int_0^3 2\pi r h \, dx$$

$$= \int_3^0 2\pi \cdot x \,(3+2x-x^2)\, dx$$

$$= \int_3^0 6\pi x + 4\pi x^2 - 2\pi x^3 \, dx$$

$$= \left[3\pi x^2 + \frac{4}{3}\pi x^3 - \frac{1}{2}\pi x^4\right]_3^0$$

$$= -\left(3\pi \cdot 3^2 + \frac{4}{3}\pi \cdot 3^3 - \frac{1}{2}\cdot\pi\cdot 3^4\right)$$

$$= \frac{81}{2}\pi = 80.11\ldots$$

$$\frac{99}{2}\pi$$

Exercises 6.3 5, 7, 11, 13, 15, 17, 19, 25(a), 39, 41, 43.